SST-225: Applied Geographic Information Systems

Professor Leif Brottem

Spatial Statistical Analysis of Urban Green Space and Environmental Justice in Seattle, WA

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Background

In addition to inequality in the distribution of environmental *hazards*, environmental justice movements often consider inequality in the distribution of environmental *amenities*. Particular focus is often on urban green space and access to the natural environment. Studies have demonstrated the importance of these spaces, showing links between green space and perceived safety [3], public health [2], and more. Additionally, studies have shown evidence of racial and/or class bias in the accessibility of urban green spaces [3].

As the distribution of urban green space is an inherently geographic issue, integration of GIS and spatial statistics into the analysis process could yield valuable results.

Methods

Research Question

Is there a significant relationship between the presence of urban green space in an area and the median yearly income of its resident population?

Data Sources and Management

Study area was limited to Seattle, WA. Data was gathered from the USGS Earth Explorer, the NHGIS database, the City of Seattle open data portal, and the ACS 2010 census. Four environmental amenity metrics—**explanatory variables**—were generated at the census block group level (data management methodologies in Figure 1)

(a) Mean Normalized Difference Vegetation Index
(b) Percentage Canopy Cover
(c) Tree Density
(d) Percentage Park

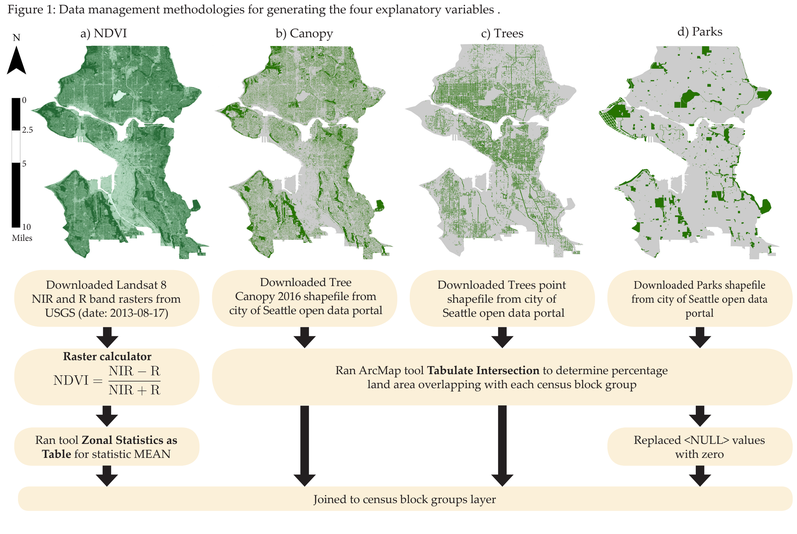
Ordinary Least Squares Regression Model

I ran an OLS regression in GeoDa with the four explanatory variables above and median income as the **dependent variable** (output in Table 2 and diagnostics in Table 1). Diagnostics indicated significant spatial autocorrelation and significance in the Robust Lagrange Multiplier (Lag) statistic, necessitating a spatial lag model [1].

Spatial Lag Regression Model

In order to account for spatial autocorrelation in the residuals, I ran a **spatial lag** regression model in GeoDa using a row-standardized queen-contiguity weights matrix. Model output is shown in Table 3 and diagnostics in Table 1.

Figure 1: Data management methodologies for generating the four explanatory variables .



Results

Preliminary data analysis showed evidence of some positive correlation between the explanatory variables and median income (Figure 2).

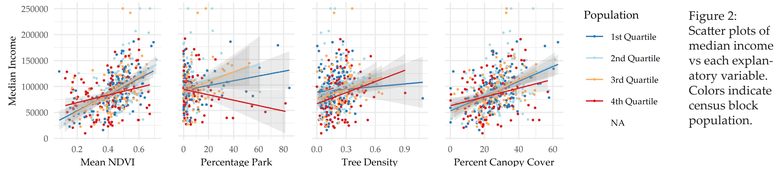


Figure 2: Scatter plots of median income vs each explanatory variable. Colors indicate census block population.

OLS regression showed an adjusted R^2 of 0.175 and significance in all variables except for Percent Park (Tables 1 and 2). However, diagnostics for spatial dependence indicated significant **spatial autocorrelation**, or geographic clustering of alike values, in the OLS residuals ($I = 0.344$, Figure 5). This introduces the possibility that individual observations could be influenced by their neighbors, violating fundamental regression assumptions [1].

The spatial lag regression model accounts for this possibility by introducing a new explanatory variable representing neighborhood relationships.

	R^2	Adj-R^2	Log likelihood	Akaike Info	Schwarz	MC#	J-B	B-P	K-B
OLS	0.182	0.175	-5947.32	11904.6	11928.7	16.15	72.70	27.91	15.53
LAG	0.434	N/A	-5875.96	11763.9	11789.2	N/A	N/A	33.74	N/A

Table 1: Regression Diagnostics

Figure 3 shows the mapped output of the spatial lag regression model:

(a) Predicted median income values
- The hypothetical distribution of median income assuming the four explanatory variables explain everything
- Any variability not accounted for in the model is removed

(b) Observed values of median income
- Included for comparison to predictions
- Directly from 2010 census

(c) Spatial lag residuals
- Indicates specific areas of over- and under-prediction of the model
- Should lack spatial autocorrelation

Variable	Coefficient	Std Error	t-Statistic	Probability
Constant	34903.76	5527.683	6.314357	0.0000000
Mean NDVI	50831.7	23696.01	2.145159	0.0324270
Percent Canopy	802.0287	315.4792	2.542255	0.0113182
Tree Density	53355.9	11938.7	4.469155	0.0000097
Percent Park	-83.71951	159.0359	-0.5264189	0.5988365

Table 2: Regression output for ordinary least squares model.

Variable	Coefficient	Std Error	t-Statistic	Probability
Lagged Income	0.6072214	0.04402005	13.7942	0.0000000
Constant	4372.366	5245.884	0.8334852	0.4045710
Mean NDVI	4372.366	5245.884	0.8334852	0.4045710
Percent Canopy	-51.11506	131.6358	-0.3883066	0.6977893
Tree Density	28422.01	19783.74	1.436635	0.1508219
Percent Park	475.7419	261.0508	1.822411	0.0683925

Table 3: Regression output for spatial lag model.

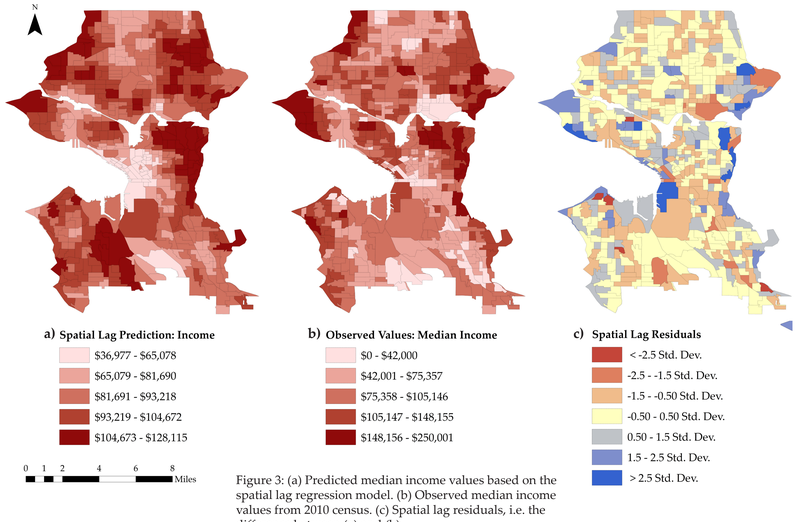


Figure 3: (a) Predicted median income values based on the spatial lag regression model. (b) Observed median income values from 2010 census. (c) Spatial lag residuals, i.e. the difference between (a) and (b).

Analysis

Comparative diagnostics indicate a significant improvement in the fit (Table 1)

- Increased R^2 from 0.182 to 0.434
- Increased Log Likelihood (+71.36)
- Decreased Akaike Info Criterion (-140.7)
- Decreased Schwarz Criterion (-139.5)

The spatial lag model changed the relative significance of variables:

- Percent park became more significant and all other variables become somewhat less significant (Tables 2 and 3)
- Implies that some of the original significance of these variables was due to geographic proximity in the observations

The spatial lag model removed nearly all spatial autocorrelation in the residuals, as it should (Figure 5).

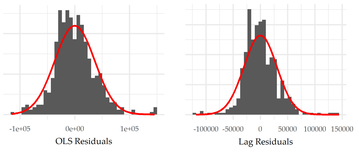


Figure 4: Histograms of OLS and spatial lag resiudals, fitted to a normal distribution.

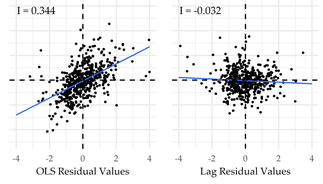


Figure 5: Moran's I scatter plots for OLS and spatial lag residuals.

OLS model

- Adjusted R^2 of 0.175 indicates fairly poor correlation
- Jarque-Bera test of 72.70 indicates poor normality of errors
- Breusch-Pagan test of ($p < 0.00005$) indicates heteroskedasticity
- Koenker-Bassett test of ($p < 0.005$) indicates heteroskedasticity

Spatial Lag Model

- Breusch-Pagan test ($p < 0.000005$) indicates that heteroskedasticity remains an issue
- Figure 4 shows histograms indicating the issue with the normality of errors in the OLS and lag models

Conclusions

The spatial lag regression model demonstrated a significant positive correlation between the four explanatory variables (NDVI, canopy cover, tree density, and percent park) and median income at the census block level. The significance of this relationship was not evident in the standard OLS regression model, as it did not account for spatial autocorrelation and the possibility that observations are related by virtue of their geographic location. Furthermore, visual inspection of Figure 3 indicates that the spatial lag model was able to predict the general distribution of income based only on the environmental variables, indicative of a larger issue at play with regards to the distribution of income and environmental amenities in urban communities.

Acknowledgements

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References

[1] Anselin, L. (2005). *Exploring spatial data with GeoDa: A workbook*. Chapter 22, pages 165–223. Santa Barbara, CA: Center for Spatially Integrated Social Science.

[2] Jennings, V., Browning, M., & Rigolon, A. (2019). *Urban green spaces: Public health and sustainability in the United States*. Cham, Switzerland: Springer.

[3] Li, X., Zhang, C., & Li. W. (2015). Does the visibility of greenery increase percieved safety in urban areas? Evidence from the place pulse 1.0 dataset. *ISPRS International Journal of Geo-Information, 4(3)*, 1166-1183. MDPI AG.